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The Emotion Engine

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Abstract

Language inevitably compresses and simplifies human emotion, causing subtle affective nuances to be lost in verbal expression. Although large language models (LLMs) can recognise and generate emotional language, most existing methods rely on discrete emotion labels or token-level control, which limits fine-grained and intuitive affective manipulation.

This project proposes an Emotion Engine that represents emotion as a continuous latent vector, rather than explicit linguistic categories. Given an input text, the system extracts its underlying emotional state as a low-dimensional emotion vector, which composes multiple overlapping affective directions. Users can directly adjust intensities of those vectors before regenerating text, that reflects the edited emotional state. Importantly, this operation is performed prior to language generation, bypassing the lossy compression inherent in verbalisation.

By leveraging recent advances in latent-space reasoning and internal activation steering in LLMs, the Emotion Engine enables controllable, pre-verbal emotion editing within a text-to-text pipeline. Human communication is always mediated by media. This project can affect all communication in the sense that it deals with feelings evoked by the media. Recent creative act with LLM prompt engineering may in fact be a role reversal, since creativity should serve before the verbalisation. By making it possible to edit the feelings that were really wanted to be conveyed, this project could bring about a more precise communication that society has never seen before.

Acknowledgments

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Contents

Chapter 1

Introduction

1.1 Words draw boundary lines across the ocean of "concepts"

When I was still in elementary school, my favourite biography in the library was Helen Keller's. Stricken by illness as a child, she lost both sight and hearing and bore the "triple hardship" of being blind, deaf, and mute, yet devoted herself to the advancement of education and welfare for the disadvantaged.

But Helen Keller's story is not merely a tale of "overcoming adversity." What is etched most vividly in my memory is the moment when she first understood the word "water." At first, Helen insisted that "cup" and "the water in it" were the same thing. However, the instant when Anne Sullivan signed "W-A-T-E-R" into her hand while letting the cold water from a pump run across Helen's palm, her world changed. When she could separate the object that flows from the container, the continuum of chaos was reshaped into an ordered world with partitions.

This scene exemplifies the essence of language. We do not grasp the external world directly as "things-in-themselves." The information flowing into our senses is, in truth, a continuum; language draws borders there and carves out units of meaning. Even the boundary between river and sea is not sharply inscribed by nature itself; it is defined for human convenience and by culture – "this far is river," "from here is sea." Words do not merely copy lines already inscribed upon the world; they are the very act of drawing those lines as needed. In other words, language is not just a label: it is the activity by which we impose boundaries within our cognition and shape the world we see. To learn words is not merely to expand vocabulary; it is to gain a way of partitioning the ocean of concepts and choosing how to look at the world. In that sense, we are, day by day, sketching new boundaries.

1.2 Verbalisation is a lossy compression

Verbalisation is an act of forcing the complex and continuous sensation into discretised vocabulary and grammar. When we perform **verbalisation**, there lurks a commonly overlooked risk; it is intrinsically **lossy**. The moment we verbalise, the fine nuances contained within raw feeling are abstracted into the fences of cognition that humanity has drawn at the level of words. Even if we choose the same term, the warmth that term carries depends on each person's embodied history, and it is eventually averaged out. The relief or pride we felt when we opened our old school yearbook for the first time in a decade, once put into text, may be flattened into a bland "how nostalgic." Moreover, lexicons and meanings for emotions differ greatly across languages and cultures**?**; terms like "anger" or "joy" do not necessarily point to the same nuances around the world. And yet humanity cannot let go of language. Even if it is a lossy compression, language remains the only standardised interface that connects the human heart to society.

1.3 Objectives

The goal of this project is not to deny language but to design a parallel circuit that supplements the components lost in the compression of verbalisation, i.e. a circuit for restoring what words leave behind. What if we could scoop up the human central cognition directly, without passing through the rendering pipeline of language, and translate it into other forms of expression? Could we transfer the lingering sensation that arises when reading a letter from your significant other into poetry or essays? Or could we improve novel translations by capturing emotions lost through differences in lexical definitions between languages? Here lies an uncharted gap in our understanding of human cognition.

This project aims to emulate a kind of human synaesthesia. In other words, we aim to map emotions across modalities. We will develop the foundational technology to extract, manipulate, and re-synthesise a **pre-verbal, intermediate representation of emotion**, and explore possible ways to steer the model output using those representations of emotion.

The final product of this project is a computational system capable of extracting, manipulating, and presenting the emotions contained in input text without having to go through the compression of verbalisation. The system focuses not on what the text superficially tells, but on the emotional state that existed immediately before the text was produced, and treats it as a continuous vector representation.

When a user enters any text, the system analyses the emotional nuances contained in that text and visualises them as a low-dimensional continuous vector. These emotion vectors are not discrete labels such as "joy" and "sadness," but are expressed as multiple overlapping emotional directions, thus preserving subtle mixed emotions and fluctuations that cannot be captured by language.

Users can intuitively manipulate this emotion code space. For example, by strengthening only a specific emotional direction and weakening another direction for the simultaneous presence of `relief` and `loneliness` in a certain sentence, it is possible to sensitively edit the emotional state that one really wanted to convey. It is important to note that this operation is not performed at the linguistic level, such as vocabulary selection or stylistic adjustment, but rather on the emotional expression itself before the language is generated.

The edited emotion code is again transformed into a linguistic expression. The text generated here is not simply a paraphrased sentence, but is the result of a re-expression that retains or intentionally transforms the emotion contained in the original sentence.

In this process, the user does not need to put their emotions into explicit words. The verbal averaging and simplification that is usually considered inevitable between emotion recognition and generation is bypassed, and the raw emotion felt by the user can be transferred to another form of expression in a more controllable manner.

This end product is neither a system for classifying and diagnosing emotions nor a tool for automatically generating emotional sentences. Rather, it is an interface that temporarily holds the emotions that people had before they put them into words as manipulatable intermediate expressions, from which new linguistic expressions can be launched. The ultimate value of this project lies in the fact that it makes it possible to return to the boundaries drawn by language and choose words again.

Chapter 2

Background

2.1 Emotion Representation in LLMs

2.1.1 Discrete Emotion Labels and Their Limitations

Early work treated emotion as a small set of discrete categories. The most influential taxonomy is Ekman's seven basic emotions: anger, contempt, disgust, enjoyment, fear, sadness, and surprise **?**.

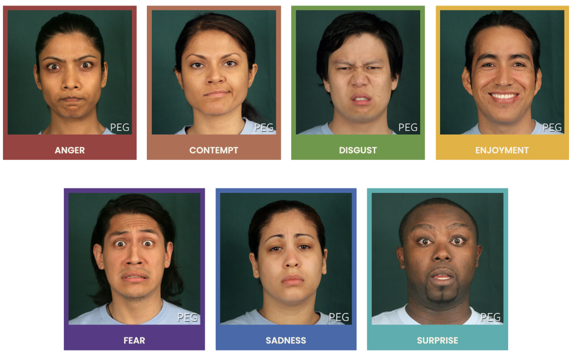

Figure 2.1: Paul Ekman's seven basic emotions **?**

This assumes that emotions are biologically universal and psychologically irreducible. This framework enabled early supervised emotion classification systems by providing a clear labeling scheme and facilitated the construction of benchmark datasets. However, the categorical assumption introduces strong inductive bias; emotions are treated as mutually exclusive and qualitatively distinct.

Subsequent datasets attempted to increase expressive granularity. GoEmotions **?**, for example, expanded the label space to 27 emotion categories and allowed multi-label annotation.

Positive		Negative		Ambiguous
admiration 👏	joy 😃	anger 😡	grief 😢	confusion 😕
amusement 😂	love ❤️	annoyance 😒	nervousness 😬	curiosity 🤔
approval 👍	optimism 🤞	disappointment	remorse 😔	realization 💡
caring 🤗	pride 😌	disapproval 👎	sadness 😞	surprise 😲
desire 😍	relief 😅	disgust 🤮		
excitement 🤩		embarrassment 😳		
gratitude 🙏		fear 😨		

Figure 2.2: GoEmotions taxonomy: 28 emotion categories including “neutral”. **?**

This enabled us to capture cases where multiple emotions occur in a single utterance. It is reported that emotions cluster by sentiment polarity and intensity, and that many categories exhibit high inter-label correlation. Nevertheless, despite its increased resolution, the dataset still relies on one-hot / sparse multi-hot encodings. Such encodings collapse emotional intensity and compositional structure into discrete indices, preventing the representation of graded transitions. For example, the difference between mild and intense anger cannot be represented, and the blend of emotions, such as bittersweet emotion, is also difficult.

Essentially, discrete labels force the learning problem into a classification regime that cannot capture continuity, geometry, or arithmetic structure. For example, two samples labeled `joy` may be emotionally distant, while `joy` and `contentment` may be emotionally close; yet this structure is invisible to the model itself. These limitations motivate a shift away from purely categorical representations toward continuous emotion spaces.

This problem of the discretised emotion labels is also raised by Wang and Zong (2021) **?**. They explicitly formalise this by arguing that one-hot emotion labels ignore latent relationships between emotion categories. They observe that emotion calsses are not orthogonal in human perception, but rather exhibit similarity in a gradation manner. More broadly, discrete labeling schemes prioritises the annotation convenience over the representational adequacy. While it simplify supervision and evaluation, they impose artificial bounadies that obscure the underlying structure of emotions. This mismatch becomes especially problematic when attempting cross-dataset transfer or cross-cultural comparison, where differences in label granularity cannot be reconciled within a purely categorical framework.

These shortcomings have motivated a growing body of work that seeks to represent emotion in continuous spaces, where distance, direction, and magnitude encode affective relationships. By having a representation methods that preserves intensity, similarity and compositionality, we can enable models to capture nuanced emotional variation beyond rigid class boundaries. This transition from discrete labels to continuous emotion spaces is the focus of the next subsection.

2.1.2 From Discrete Labels to Continuous Emotion Spaces

Continuous emotion representations seek to encode emotion as points in a low-dimensional vector space, where distance, direction, and magnitude correspond to psychological relationships. Early psychological models such as the pleasure–arousal–dominance (PAD) framework **?** formalised emotion along continuous axes, suggesting that emotions differ quantitatively rather than categorically.

1. **Pleasure** represents the hedonic quality of an experience, ranging from unpleasant to pleasant. States such as joy, contentment, and satisfaction occupy the positive end of the pleasure axis, whereas sadness, disgust, and despair lie toward the negative end.

2. **Arousal** captures the degree of physiological and cognitive activation associated with an emotional state, ranging from calm or lethargic to excited or agitated. High-arousal states include fear, anger, and excitement, while low-arousal states include relaxation, boredom, or melancholy.

3. **Dominance** reflects the extent to which an individual feels in control of, or overpowered by, a situation. High-dominance states are associated with feelings of agency, confidence, and control (e.g. pride, anger), whereas low-dominance states correspond to submission, helplessness, or vulnerability (e.g. fear, shame).

Together, these three dimensions define a continuous affective space in which canonical emotion categories correspond to regions rather than points. For example, anger and fear are both negatively valenced and highly aroused, but are separable along the dominance axis; sadness and boredom share low arousal and negative valence, but differ in dominance and intensity. This geometric interpretation naturally accounts for graded emotional intensity, mixed emotions, and smooth transitions between affective states – phenomena that are difficult to represent with discrete labels.

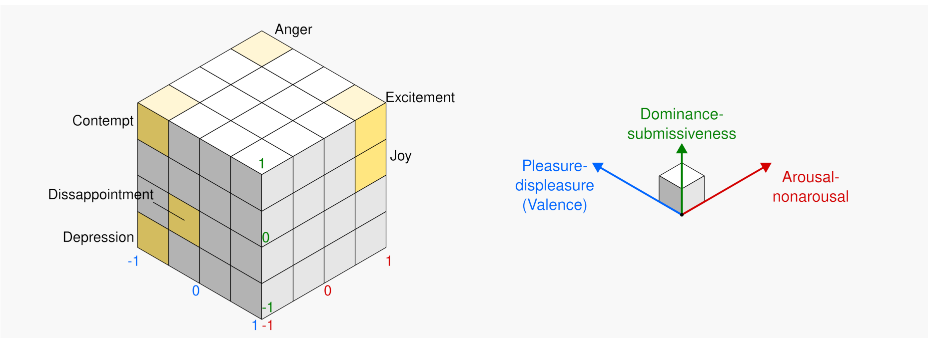

Figure 2.3: Representation of PAD model done with cubes **?**

Additionally, in the field of affective computing, an explicit attempt to bridge the gap between discrete labels and continuous emotion representations is presented by

Buechel et al. **?**. Rather than committing to a single emotion taxonomy, they propose learning *label-agnostic emotion embeddings* that serve as a shared latent representation across heterogeneous annotation schemes, including polarity, basic emotion categories, and affective dimensions such as valence and arousal.

Crucially, emotion labels in this framework are not treated as targets to be predicted directly, but as projections from a common embedding space. Models are trained to map text into this latent affective space, from which multiple label formats can be recovered via lightweight prediction heads. This design enables the learned emotion information to be transferred to another without retraining the full model. Empirically, Buechel et al. show that such embeddings preserve prediction quality while supporting cross-dataset generalisation.

Complementary to this, Wang and Zong (2021) **?** focus on learning continuous representations for *emotion categories themselves*. Instead of representing emotion classes as one-hot vectors, they derive distributed emotion embeddings by averaging soft classifier outputs over all instances of a given emotion category. These representations form an emotion space in which distances reflect empirical similarity between emotions as expressed in text. Their results demonstrate that emotion categories cluster by sentiment polarity and intensity, which captures the emotions more faithfully than the conventional semantic word embedding spaces.

Both of those approaches we looked at enable emotion to be represented as a continuous, low-dimensional structure where each of them corresponds to a region or a direction in a metric space. This perspective aligns closely with psychological theories such as PAD, while remaining grounded in data-driven learning from natural language.

Importantly, representing emotion in continuous spaces shifts the learning problem away from rigid classification toward geometric representation learning. This not only resolves many of the limitations inherent in discrete labels, but also provides a foundation for analysing how emotional information is encoded in vector representations more generally. In particular, it raises the question of whether emotion occupies identifiable subspaces or linear directions within learned embeddings, a topic explored in the following subsection.

2.1.3 Linear Structure of Emotion in Word Embedding Spaces

An important question here is whether the continuous space for emotion is actually reflected in learned language representations. Interestingly, prior to the emergence of LLMs, static word embeddings already exhibited evidence of latent affective organisation, despite being trained without explicit emotional supervision.

Wu and Jiang **?** demonstrate that emotional information in word embeddings such as GloVe **?** is not uniformly distributed across dimensions, but instead concentrated along a small number of linear directions. They train a linear autoregressive model to predict continuous emotional valence on annotated narrative data, and show that only a limited subset of embedding dimensions contributes to emotional valence.

Interpreting the learned weights reveals that emotional polarity can be captured by a low-dimensional subspace embedded within the full semantic space.

It is also reported that projecting word embeddings into this learned emotion subspace produces a substantially clearer separation between positive and nefative emotion. This indicates that emotion is linearly recoverable even when embeddings are trained without affective objectives. Namely that, emotional meaning is encoded **geometrically**, rather than a byproduct of lexical associations.

A particularly significant finding is that this projected emotion space preserves *linear compositional structure*. Vector arithmetic over words that contain emotion yields composite affective states, which is also consistent with psychological theories of emotion composition such as Plutchik's wheel of emotions **?**.

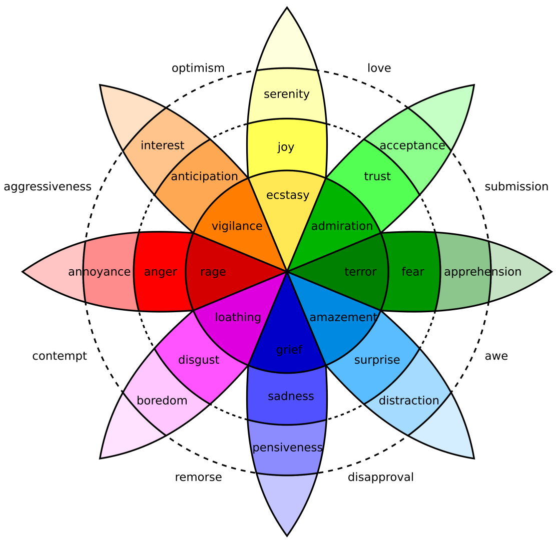

Figure 2.4: Plutchik's wheel of emotions: basic emotions are arranged radially, with angular proximity indicating similarity and radial intensity indicating emotional strength.

For example, the vector sum of `joy` and `trust` aligns closely with `love`, while remaining distant from affectively opposing states such as `remorse`. This behaviour cannot be captured by categorical representations, but emerges naturally in a continuous, vector-valued framework.

These results indicate that emotion in word embedding spaces is not only continuous but also approximately linear, admitting meaningful projections, directions, and arithmetic operations. As a result, affective similarity, intensity, and composition can be expressed through simple linear operations rather than discrete class boundaries.

Taken together, these findings provide early evidence that emotion is encoded as structured geometry within learned language representations. This establishes a representational bridge between psychological models of emotion and mathematical embedding paradigms. This also motivates further research into how such linear emotional structure scales and evolves in more expressive representation learning systems.

2.2 Manipulation of Latent Representations

To harness these emotion vectors, researchers have explored techniques for directly manipulating internal representations of models.

2.2.1 Neuron-Level Emotion Control

One line of work focuses on exploiting localised affective representations at the neuron level. For instance, Radford et al. (2017) **?** noted a single "sentiment neuron" in a byte-level recurrent language model. It has shown that the activation strongly correlates with review polarity and the manipulation of that sentiment neuron directly controls the generated sentiment. Later work shows that modern LLMs contain clusters of neurons selectively responsive to specific emotions.

Ablation and masking experiments were done by Lee et al. (2025) **?**, and it demonstrates that removing these neurons can selectively impair recognition of particular emotions. This confirms that they play a functional role rather than being incidental correlates. However, there is still a possibility that the model is redundant. For some emotions, performance remains stable after ablation, suggesting overlapping representations and compensatory mechanisms.

Additionally, low-resource fine-tuning methods such as LoRA (Low-Rank Adaptation) **?** and prefix-tuning operate at a higher level of abstraction but still manipulate latent activations. By injecting small learned vectors into the model we can induce a consistent emotional tone without full retraining. This shows that affective style is encoded in directions that are easy to steer with low-rank updates.

2.2.2 Emergent Emotional Subspaces in Large Language Models

Large language models substantially extend and generalise the affective structure previously observed in static word embeddings (2.1.3). Even though models are trained on massive corpora without any emotional information supervision, LLMs acquire rich contextual representations. These models appear to internalise emotion as a continuous and distributed property of their hidden-state geometry.

Recent work provides strong empirical evidence for this claim. Reichman et al. (2025) **?** show that emotional information in large decoder-only LLMs is organised within a low-dimensional **emotional manifold** embedded in the activation.

They analysed hidden states across layers, demonstrated how affective content aligns with a small number of stable latent directions. Importantly, these directions appeared without any explicit emotion supervision, indicating that affective structure is a natural consequence of large-scale language modelling rather than a task-specific artefact. Moreover, the identified emotional subspace is remarkably consistent across domains and languages, suggesting that LLMs internalise a broadly universal affective geometry.

Crucially, this geometry is not merely descriptive. Reichman et al. **?** show that small interventions along these latent directions can alter the model's emotional interpretation of an input, while preserving its semantic content. This establishes emotional subspaces as functionally meaningful components of the model's representation space, rather than passive correlates of surface-level sentiment cues. This establishes emotional subspaces to be considered as a functional and meaningful components of the model's representation space.

Complementary evidence comes from neuron-level analyses too. Lee et al. (2025) **?** identify groups of neurons whose activations selectively track specific basic emotions in large instruction-tuned models. Emotional information is broadly distributed, but certain neurons constantly exhibit disproportionately strong responses to particular affective states. Further ablation experiments revealed that suppressing these neurons can significantly degrade emotion recognition performance for some emotions. This provides a causal evidence that emotional information is actively processed rather than epiphenomenal.

Taken together, these findings suggest that emotion in LLMs is best characterised as an emergent, low-dimensional subspace embedded within high-dimensional contextual representations. Emotional meaning is not localised to single units, nor reducible to discrete labels; instead, it occupies a structured region of stable latent space. This emergent affective geometry provides the representational foundation that enables direct control of emotional behaviour through linear interventions, which we discuss next.

2.2.3 Steering Latent States with Linear and Continuous Control Vectors

If emotional meaning in LLMs is encoded as directions or subspaces within latent representations, a natural consequence is that affective behaviour should be directly controllable by intervening along those directions. This observation has motivated a line of work on **latent steering**, in which linear and continuous control vectors are applied directly to internal activations during inference.

An early and influential approach is Plug-and-Play Language Models (PPLM) **?**, combines a frozen language model with an external attribute model (e.g. a sentiment classifier) and performs gradient ascent on the hidden states to increase the likelihood of a desired attribute. Effectively, this method introduces a small **sentiment vector** into the latent representation, biasing generation toward positive or negative emotion without modifying the model's weights. Although originally framed as a controllable generation technique, PPLM implicitly relies on the assumption that affective attributes correspond to approximately linear directions in latent space.

More recent work makes this assumption explicit and removes the need for iterative optimisation. Anthropic's **Contrastive Activation Addition** (CAA) **?** directly intervenes in a model's internal activations using precomputed **steering vectors**. These vectors are obtained by collecting large sets of positive and negative examples of a target behaviour or attribute and averaging the differences in their intermediate representations. The resulting vector captures a direction in latent space that causally corresponds to the desired attribute. During inference, adding this vector to the residual stream steers model behaviour in a predictable manner, while leaving the model parameters unchanged.

A key advantage of CAA is its simplicity and interpretability. The intervention is linear, and the strength of the effect can be modulated continuously by scaling the steering vector with a scalar coefficient. Moreover, multiple steering vectors can be combined additively, enabling compositional control over several attributes simultaneously. In contrast to prompt engineering or fine-tuning, such interventions operate directly on the model's internal state and often yield more precise and robust control, particularly for abstract properties such as sentiment or emotional tone.

Crucially, steering latent states does not only affect surface-level lexical choices. Injecting affective control vectors at intermediate layers can alter the model's internal interpretation of emotional context, changing how emotional cues are integrated and propagated through the network. This supports the view that emotion in LLMs functions as a global latent variable, rather than as a property tied to individual tokens or outputs.

As conclusion, latent steering methods such as PPLM and CAA provide strong evidence that emotional subspaces in LLMs are not only emergent and structured, but also causally actionable. Manipulating affective behaviour through simple linear interventions reinforces the interpretation of emotion as a continuous geometric representation space. It further bridges descriptive analyses of emotional geometry with concrete mechanisms for controlling model behaviour.

2.2.4 Causal Evidence for Linear Sentiment and Emotion Directions

The existence of affective directions in representation space can be shown by linear probes or dimension reduction, however, it is still doubtful if they actually play a causal role in model behaviour. Recent work provides strong evidence that emotion directions in LLMs are not only linearly recoverable, but also functionally necessary and sufficient for affective reasoning.

Hollinsworth et al. (2024) **?** show that sentiment in LLMs is encoded along a single dominant linear direction in contextual embeddings. They perform directional ablations and interventions, demonstrating that removing / suppressing this sentiment direction causes systematic degradation in downstream emotion-sensitive tasks such as emotion classification. Conversely, injecting the same direction into neutral contexts induces predictable shifts in model outputs. This bidirectional manipulation provides an evidence that the emotion direction is actively used by the model rather than being a passive correlate.

Importantly, affective competence in LLMs does not depend on explicit fine-tuning for emotion-related tasks. Broekens et al. (2023) **?** demonstrate that off-the-shelf

LLMs such as ChatGPT can map text to precise valence–arousal–dominance (VAD) values and detect fine-grained emotional cues using prompting alone. This sugegsts that continuous affective representations are already embedded in the latent space as part of general language understanding. This is a strong evidence that interpretes emotion as a pre-verbal latent variable, rather than a task-specific output format.

At larger scales, evidence further indicates that emotional representations are not flat but hierarchically structured. Zhao et al. (2024) **?** observe that very large models (more than 405 billion parameters) develop tree-like emotion representations, where basic affective states branch into more specific emotions. Models featuring a more consistent hierarchical affective geometry demonstrate enhanced emotion recognition capabilities and increased persuasive effectiveness in negotiation scenarios. This directly links the internal structure of emotional expression to functional behaviour.

These findings converge on a causal interpretation of affective geometry in LLMs. Removing emotion directions impairs affective reasoning, while injecting them predictably alters behaviour. Moreover, emotional information is not confined to isolated tokens or neurons, but is accumulated and summarised across layers into global latent variables that shape general affective capabilities of models.

In conclusion, there is a principled foundation for vector-based manipulation of emotion, which enables us to express emotional nuance through numeric operations on embeddings rather than token-level substitutions. This causal grounding is essential for our project that seeks to control and align emotional behaviour in large language models.

2.3 Reasoning Beyond Token Space

2.3.1 Limitations of Token-Based Reasoning

Conventional LLMs reason in token space. Because inputs are projected onto the discrete boundaries humanity drew as "vocabulary," fine affective nuance and embodied subtleties are irretrievably lost during that projection, and what remains in one output language may differ from what remains in another. To adjust tone, one often uses prompt engineering or instruction tuning ("Please answer politely," etc.), but adherence is uncertain and fine-grained control is difficult. Other countermeasures include supervised fine-tuning to internalise a style, or label-conditioned generation (e.g. adding `<joy>` `<sad>`), but fine-tuning is costly and switching between styles is inflexible; label conditioning also cannot capture complex blends of nuance.

Beyond affective control, the same limitation applies to reasoning itself. Autoregressive language models do not take token functionality into account in its generation process. Even if the token carries significant reasoning content, it will be allocated approximately same computational budget as grammartical tokens like `a` or `is`. As a result, only a small subset of internal states carries the actual inferential work, and most of the tokens in explicit reasoning traces do not contribute to the underlying computation.

Recent works show this disconnectivity between the surface output tokens and internal reasoning process. Pfau et al. (2024) **?** demonstrate that transformers can perform non-trivial hidden computation even when output tokens are semantically uninformative. For example, inserting filler or pause tokens (carry no explicit meaning) somehow measurably improve performance on reasoning tasks. This suggests that the essential computation is carried by latent activations rather than by the emitted tokens themselves. Token generation only a communicative language constraint around an internal process.

The inefficiency of token-based reasoning becomes more apparent when reasoning is explicitly verbalised. Chain-of-thought prompting improves performance in many tasks, however **it requires the model to externalise intermediate states into natural language, even when those states are not naturally linguistic**. As argued by Hao et al. (2025) **?**, language is optimised for communication rather than for computation. Forcing reasoning to unfold within the constraints of a discrete vocabulary introduces unnecessary overhead and distortion. Many reasoning tokens exist not because they are required for inference, but because they are required for human readability.

From this perspective, explicit verbalisation constitutes a form of lossy compression for an affective reasoning as discussed in the previous section. The continuous and high-dimensional internal states during inference must be projected onto a discrete and culturally contingent token space, which flattens distinctions that are visible for computation but words-level invisible. This compression is problematic for affective computing, because emotion structure is inherently graded, which is resistant to precise lexicalisation.

Taken together, these observations suggest that token space not a substrate of reasoning, but rather an interface. While it provides a convenient medium for communication and supervision, it is suboptimal for internal computation. This motivates a growing line of work that seeks to decouple reasoning from explicit token generation. Computation can proceed without the constraints imposed by language.

2.3.2 Reasoning in Continuous Latent Space

Motivated by the observation that reasoning is primarily carried by latent activations rather than surface tokens, a growing body of work has explored **implicit** forms of chain-of-thought. The central idea is to retain the intermediate reasoning within the model's hidden state, so that we can benefit from the multi-step reasoning while avoiding the distortions introduced by forcing intermediate states into natural language.

Early implicit chain-of-thought approaches aim to internalise reasoning traces by gradually removing explicit reasoning tokens during training, encouraging the model to perform the same computation silently in its latent space. More recent methods go further by explicitly allocating computation to continuous latent representations that replace textual reasoning steps altogether. In these models, reasoning unfolds as a sequence of hidden-state transitions, while the output layer is used only to produce the final answer.

A representative work is **CODI** (Compressing Chain-of-Thought into Continuous Space via Self-Distillation) **?**. This approach compresses explicit chain-of-thought trajectories into a compact sequence of continuous latent states, and by distilling a teacher model that produces explicit reasoning traces, CODI trains a student model to reproduce the same reasoning outcome using a shorter sequence of latent vectors. This demonstrates that the information carried by textual reasoning can be encoded more efficiently in continuous space, without requiring token-level supervision at inference time.

A more radical approach is proposed by **Coconut** (Chain of Continuous Thought) **?**. Rather than compressing explicit reasoning traces, Coconut modifies the inference procedure itself by feeding the model's last hidden state back as the next input embedding. In this way, reasoning unfolds entirely within continuous latent space and there is no need of intermediate textual tokens at all. However, because Coconut relies primarily on answer-level supervision, the resulting latent trajectories can be unstable when extended to deeper reasoning chains, which we discuss in the next subsection.

2.3.3 Implicit and Continuous Chain-of-Thought

However, there is one challenge that purely implicit reasoning introduces. When multiple latent reasoning steps are chained together without sufficient structural constraints, the latent representations can collapse into homogeneous states. This phenomenon is often referred to as **latent collapse**, which undermines the model's ability to maintain distinct intermediate representations.

SIM-CoT (Supervised Implicit Chain-of-Thought) **?** addresses this limitation by introducing step-level supervision during training. Instead of supervising only the final answer or the overall reasoning trajectory, SIM-CoT attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase. This alignment ensures that each latent vector encodes distinct, semantically meaningful content, preventing homogenisation and improving stability. Crucially, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision. Empirically, SIM-CoT substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style architectures.

It attaches an auxiliary decoder that forces each latent reasoning step to predict a corresponding explicit reasoning phrase, which ensures that each latent vector to encoded semantically meaningful content. This substantially improves both performance and robustness over earlier latent reasoning models, including Coconut-style architectures. Importantly, the auxiliary decoder is used only during training and is removed at inference, allowing the model to retain the efficiency of silent latent reasoning while benefiting from structured supervision.

These approaches demonstrate that chain-of-thought need not be realised as a sequence of discrete tokens. Reasoning can be instead implemented as a controlled trajectory through continuous latent space, with explicit language serving primarily as a training scaffold or interpretability aid. Implicit and continuous chain-of-thought models provide a natural foundation for studying how emotions may be represented and manipulated within latent reasoning processes.

2.3.4 Implications for Affective Representation

The developments reviewed above suggest a converging picture of representation and computation in LLMs. As discussed in earlier sections, emotions are not confined to discrete labels or lexical choices but it rather emerges as a continuous structure within latent space. At the same time, recent advances in implicit chain-of-thought demonstrate that reasoning itself can be implemented as a trajectory through that same latent space, without requiring intermediate symbolic tokens.

These findings summarises to the fact that emotion need not be treated as separate components at the level of surface language. Instead, it appears to be grounded in shared latent representations that evolve over the course of inference. From this perspective, emotional information may constitute an integral part of the internal reasoning state itself, shaping how intermediate representations are formed and propagated.

Importantly, this possibility remains largely unexplored. Although latent reasoning frameworks have demonstrated improved eff expressivity for logical and mathematical tasks, their implications for affective representation have not been systematically examined. Whether emotional dimensions align or interact with latent reasoning trajectories is an open question. Addressing this gap requires moving beyond token-level analysis and considering emotion as a first-class component of latent computation rather than as a stylistic attribute.

These considerations motivate the investigation in the following chapters. Rather than treating emotion as an external label or an output constraint, we explore the hypothesis that affective states can be represented within continuous latent spaces that underlie reasoning itself. By doing so, we aim to better understand how emotional meaning may be re-expressed when language is no longer the primary medium of inference.

Chapter 3

Development

This chapter translates the conceptual claims of the previous chapter into an empirical pipeline.

Our central objective is to test whether emotion can be recovered and steered in residual-space representations of a frozen language model. To achieve this, we progress from controlled data construction and residual-stream extraction to layer diagnostics, linear decodability analysis and causal steering via activation-space directions. We then examine the latent geometry that underpins these processes and evaluate whether steering can produce directional, monotonic affective change while preserving semantic content. In short, this chapter not only asks whether emotion is represented in hidden states, but also whether this representation can be used to control emotional editing.

3.1 Dataset Construction

In order to investigate whether emotion is encoded as structured geometry in residual space, we constructed controlled rewrite datasets from shared source utterances. The primary dataset is an emotion-category rewrite set derived from two dialogue corpora: DailyDialog **?** and EmpatheticDialogues **?**. We selected dialogue data because conversational utterances provide compact emotional signals grounded in context in everyday interpersonal situations, while keeping propositional content sufficiently constrained for controlled rewriting. For each source utterance, we used GPT-4.1-mini to generate rewrites for Plutchik's eight basic emotions (joy, trust, fear, surprise, sadness, disgust, anger and anticipation) at three matched intensity levels (low, medium and high). This yields $8 \times 3 = 24$ affective variants per source text. A central constraint throughout the construction process was to preserve meaning: the underlying situation and factual content were held constant while only the affective framing was systematically varied.

Emotion	Utterance
Base	**I waited for hours and nobody came.**
Joy	Nobody came for hours, but I ended up enjoying the quiet time on my own.
Trust	Nobody came for hours, but I am sure they had a valid reason.
Fear	Nobody came for hours, and I started worrying that something might have gone wrong.
Surprise	Nobody came for hours, and honestly I did not expect that at all.
Sadness	For hours, nobody came. I guess that says enough.
Disgust	Nobody came for hours; that felt deeply inconsiderate.
Anger	Nobody came for hours, and I am really frustrated about wasting all that time.
Anticipation	Nobody came for hours, but I kept thinking they might still arrive any minute.

Table 3.1: Illustrative rewrites for all eight Plutchik emotion categories at matched medium intensity.

As a control condition, we constructed an additional neutral paraphrase baseline dataset. In this dataset, each source utterance has been rewritten into an affectively neutral but semantically equivalent form. This baseline provides a reference representation of content without targeted emotional colouring, enabling contrastive analysis between emotion-conditioned and emotion-suppressed activations.

Field	Example
Base utterance	I waited for hours and nobody came.
Neutral paraphrase	I waited for several hours, and no one arrived.
Purpose	Preserve events and meaning while removing affective phrasing.

Table 3.2: Illustrative example from the neutral-paraphrase control dataset.

Together, these datasets enable us to distinguish emotional signals from lexical-semantic content. This provides an empirical basis for the decodability, geometry and steering analyses presented in subsequent sections.

This design makes it possible to ask whether affective variation forms a consistent direction in activation space, rather than merely reflecting surface-level lexical differences between emotional words.

3.2 Residual-Stream Extraction

We extract hidden-state activations from a frozen, instruction-tuned language model (Llama-3.1-8B **?**) with no gradient updates. Our aim is to determine where affective information is represented in the model's internal computations rather than in output tokens alone. The residual stream is therefore used as the primary analysis target because it is the shared channel through which each transformer block accumulates and propagates semantic and affective information across layers.

To ensure that the inputs are comparable across the different conditions, each sample is formatted with a shared instruction prefix that is constructed from the same base utterance. Only the continuation text is changed, either through an emotion rewrite or a neutral paraphrase. Specifically, we use a user instruction in the form of a chat template that asks for a rewrite aligned with the speaker's feelings, and then append the rewritten sentence as the assistant continuation. This matching protocol controls for variance at the prompt level and enables direct comparison of emotion-conditioned and neutral-conditioned activations.

Template Prompt
Please rewrite this so it aligns with my feelings more. Start straight from the rewritten text without any preamble, and only provide **one** rewritten version. Base: {`base_text`}

Table 3.3: Prompt format used for activation extraction.

Residual sampling involves forwarding batched inputs with hidden-state output enabled and collecting vectors at selected hook layers. For each target layer l, the last token's hidden state is taken from the layer's output, i.e.

$$\mathbf{h}_l = \texttt{hidden\_states}[l+1][:, -1, :]$$

Since tokenisation uses left padding, the index -1 consistently corresponds to the final real token in each sequence in the batch. Final-token representaion is the state after the model has processed the entire rewritten utterance, making it a compact summary of the affective framing expressed by the continuation. Activations are sampled from layers $\{8, 10, 13, 16, 19, 22\}$, producing a multi-layer representation for every input.

The emotion-intensity extraction pipeline produces a 5D tensor:

$$\mathbf{H}^{\text{emo}} \in \mathbb{R}^{N \times E \times I \times L \times D},$$

where N is the number of source texts, $E = 8$ emotions, $I = 3$ intensity levels, L is the number of sampled layers, and D is the hidden dimension. The neutral baseline pipeline produces:

$$\mathbf{H}^{\text{neu}} \in \mathbb{R}^{N \times L \times D}.$$

Source IDs are sorted to align both tensors by index, allowing direct subtraction

$$\Delta \mathbf{H}_{n,e,i,l,:} = \mathbf{H}^{\text{emo}}_{n,e,i,l,:} - \mathbf{H}^{\text{neu}}_{n,l,:},$$

which isolates affective shifts while controlling for base semantic content.

3.3 Layer Selection and Diagnostics

Not all transformer layers provide equally useful representations for affective analysis. Emotion-related structure may be present at multiple depths, but different layers can privilege different properties of the representation. For example, one layer may provide strong emotion-category separability, while another may better preserve a compact low-dimensional geometry. Since the later analyses require both linear decodability and geometrically stable affective directions, we compared candidate layers using several complementary diagnostics.

Based on these diagnostics, layer 13 was selected as the primary layer for the main analyses. It is not globally best on every metric, but it provides the most convincing overall trade-off between category separability, low-dimensional compactness, and sensitivity to graded affective variation.

Layer	Emotion probe BAcc *(4-D subspace)*	Centroid EVR_4	Mean local PC1 ρ
10	0.6751	0.8876	0.0998
13 *	**0.7033**	0.8920	**0.1336**
16	0.6704	**0.8958**	0.1297

Table 3.4: Layer-wise diagnostic summary for candidate layers. Emotion-category probe balanced accuracy (4-D subspace), centroid compactness (EVR_4), and mean local PC1 intensity correlation. Bold indicates the best value per column; asterisk marks the selected reporting layer.

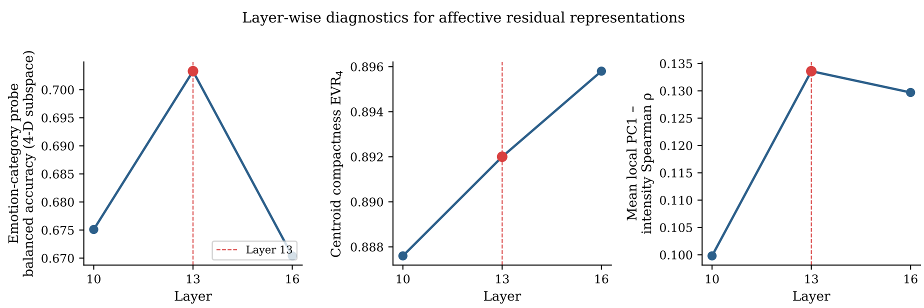

Figure 3.1: A visualisation of the layer diagnostics across candidate layers.

First, emotion-category probe performance in the 4D affective subspace is highest at layer 13. The balanced accuracy reaches 0.7033 at layer 13, compared with 0.6751 at layer 10 and 0.6704 at layer 16. On this criterion, layer 13 preserves the most linearly decodable information about the target emotion categories.

Second, this increase in decodability does not come at the cost of geometric compactness. The centroid-level four-dimensional explained variance remains high at layer 13, with $\text{EVR}_4 = 0.8920$. This is slightly below layer 16 ($\text{EVR}_4 = 0.8958$)

but above layer 10 ($\mathrm{EVR}_4 = 0.8876$). In other words, layer 13 is not the strongest layer on this metric alone, but it remains close to the best-performing depth while retaining stronger emotion-category separability.

Third, layer 13 shows the strongest sensitivity to graded affective variation among the candidate layers. The mean local PC1 intensity correlation is 0.1336 at layer 13, compared with 0.0998 at layer 10 and 0.1297 at layer 16. Although the absolute correlation is modest, the relative increase suggests that layer 13 is the most sensitive of the three candidate layers to systematic changes in affective intensity.

To avoid overfitting the analysis to a single depth, we also examined neighbouring layers as robustness checks. Layers 10 and 16 show similar qualitative trends and preserve the main geometric findings, suggesting that affective structure is not confined to a single transformer block. We therefore do not treat layer 13 as uniquely optimal in a global sense. Rather, we use it as the main reporting layer because it offers the best trade-off across the three diagnostics, while layers 10 and 16 serve as nearby robustness checks.

3.4 Linear Probe Analysis

3.5 CAA Vector Construction

3.6 Latent Geometry of Emotion

3.7 Steering Experiments

3.8 Limitations and Transition

The analyses in this chapter are intended to establish four claims: that CAA directions can be constructed from the emotion rewrite dataset, that emotion is reasonably well separated in a narrow band around layer 13, that the resulting space contains both common emotionalisation structure and emotion-specific residual structure, and that steering can satisfy directionality, approximate monotonicity, and partial semantic preservation. The next chapter turns these representational claims into an explicit evaluation protocol, covering controlled steering benchmarks and human-centred assessment.

Chapter 4

Evaluation Design

This chapter describes how the steering behaviours identified in the previous chapter will be evaluated in a controlled and human-centred manner. The goal here is not to re-establish the existence of affective structure in residual space, but to assess whether the extracted control directions yield perceptible, monotonic, and semantically stable emotional edits.

4.1 Evaluation Setup

We prepare a set of input texts spanning neutral descriptions, short narratives and conversational utterances. For each input, the system first extracts a baseline emotion vector, which is then modified along one or more predefined directions.

For controlled experiments, we consider three conditions:

1. no steering (baseline)
2. single dimensional manipulation within the PAD space, where users adjust pleasure, arousal, or dominance
3. two-dimensional manipulations within the PAD space

We define three primary metrics to evaluate the behaviour of the emotion code:

1. **Shift Accuracy**: the proportion of cases in which human annotators agree that the steered output expresses a stronger presence of the target emotion than the baseline
2. **Intensity Monotonicity**: evaluates whether increasing the magnitude of a steering vector leads to a monotonic increase in perceived emotional intensity. This metric directly tests whether emotion behaves as a continuous and scalable latent variable.

3. **Content Stability**: whether semantic content is preserved under emotional steering. This is measured using a combination of human judgement and embedding-based semantic similarity in concept space.

4.2 Human-Centred Evaluation

In addition to controlled steering experiments, we would like to conduct a small-scale user study focusing on interactive manipulation of emotion vectors. Participants are presented with the PAD visualisation and are asked to modify the emotional state of a given input text to match an intended affective description (e.g. “more tense but less negative”).

We evaluate whether users can reliably reach similar regions of the PAD space for similar intentions, whether users report that changes in the output align with their intuitive expectations, and whether fine-grained adjustments lead to perceptible but non-disruptive changes in tone. This evaluation focuses on intuitive usability rather than task performance, reflecting the project’s goal of enabling pre-verbal emotional editing rather than emotion classification.

4.3 Failure Cases and Limitations

For the cases where steering fails to produce the intended emotional shift or results in semantic distortion, we would like to analyse how those happened. It might be the case that the steering magnitudes are too high for the humans to recognise the differences, or the interference between emotion dimensions. Such cases are not treated as errors but as signals of the limits of linear emotion composition and of the PAD projection itself, hence better ways of representing the latent emotion representation should be further investigated.

Chapter 5

Evaluation Setup

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Chapter 6

Implementation and Preliminary Experiments

This chapter describes the first phase of implementation, which focuses on adapting the Coconut latent reasoning architecture to the affective regression task and diagnosing the representational capacity of the resulting latent space.

6.1 Dataset

All experiments use the **EmoBank** corpus **?**, a sentence-level dataset annotated with continuous Valence–Arousal–Dominance (VAD) scores on a scale from 1 to 5. The dataset was split into 7,801 training samples and 861 test samples. Each training instance consists of an input sentence paired with a gold PAD triple $\{V, A, D\} \in [1, 5]^3$. Reasoning-step annotations were generated offline using GPT-4o and cached as a `JSONL` file, providing a chain-of-thought scaffold for the staged training curriculum described below.

6.2 Model Variants

Four model variants were trained and evaluated, all using GPT-2 (117M parameters) **?** as the backbone.

CoT baseline. A standard chain-of-thought model in which each training example is formatted as a free-text reasoning trace followed by the gold PAD answer. The model is supervised with cross-entropy on the full token sequence.

Coconut baseline. The original Coconut architecture **?** replaces textual reasoning tokens with a fixed number of *latent tokens*. At each training stage k, the first k

reasoning steps are replaced by latent embeddings that are filled in with the model's own hidden states in a multi-pass forward procedure. The number of latent tokens per stage is controlled by the hyperparameter $c_{\text{thought}} = 2$, and up to $k_{\max} = 5$ stages are scheduled.

CoconutGPT-SimCoT. Extends the Coconut baseline with an auxiliary explainability head that, during training only, decodes each latent reasoning step into a natural-language phrase. This is analogous to the SIM-CoT supervision strategy **?** and is intended to prevent latent collapse by ensuring each latent vector encodes semantically distinct content.

CoconutGPT-Factored. A new architecture introduced in this project. Two auxiliary losses are added on top of the Coconut latent pass:

1. **VAD regression loss.** A linear head $\mathbf{W}_{\text{vad}} \in \mathbb{R}^{3\times d}$ maps the mean-pooled latent representations to a predicted PAD triple $\hat{\mathbf{v}} \in \mathbb{R}^3$. The regression loss is:

$$\mathcal{L}_{\text{reg}} = \frac{1}{3}\|\hat{\mathbf{v}} - \mathbf{v}^*\|_2^2$$

   where $\mathbf{v}^* = (V^*, A^*, D^*)$ is the gold annotation.

2. **Contrastive geometric loss.** Within each mini-batch of size B, the cosine similarity matrix of mean-pooled latent vectors is aligned to a target similarity derived from PAD-space Euclidean distance:

$$s^*_{ij} = \exp\left(-\frac{\|\mathbf{v}^*_i - \mathbf{v}^*_j\|_2}{\tau}\right), \quad i \neq j$$

   The contrastive loss is the mean squared error between the computed cosine similarities and s^*_{ij} over all off-diagonal pairs.

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}} + \lambda_{\text{con}}\,\mathcal{L}_{\text{con}}$$

with $\lambda_{\text{reg}} = 1.0$, $\lambda_{\text{con}} = 0.1$, and $\tau = 1.0$. The model is warm-started from the Coconut baseline checkpoint.

6.3 Phase 1: Linear Probe Diagnosis

Before training CoconutGPT-Factored, we conducted a linear probing experiment to determine whether PAD information was already linearly decodable from the Coconut baseline's latent representations.

For each sample, a forward pass was performed through the Coconut baseline (checkpoint at epoch 25). The mean-pooled hidden states at all latent token positions were extracted as a 768-dimensional vector. A Ridge regression probe was then trained on the 7,801 training representations and evaluated on the 861 test representations independently for each PAD dimension.

Dimension	**Pearson r (test)**	**R^2 (test)**	**R^2 (train)**
Valence (V)	0.695	0.474	0.860
Arousal (A)	0.371	-0.000	0.804
Dominance (D)	0.409	0.101	0.676

Table 6.1: Linear probe results on the Coconut baseline latent representations. Arousal shows near-zero test R^2, indicating that the information is not linearly encoded in the baseline latent space.

The results reveal a clear asymmetry: Valence is moderately recoverable (Pearson $r = 0.695$), consistent with the hypothesis that surface-level lexical sentiment is already encoded in the latent representations. By contrast, Arousal achieves a test R^2 of -0.000, which is below the constant-mean baseline, and Dominance achieves only $R^2 = 0.101$. The large gap between training and test R^2 across all dimensions indicates that the latent space is high-dimensional relative to the amount of affective information it encodes, leading to over-fitting in the probe even with regularisation.

This diagnostic confirms the central hypothesis motivating CoconutGPT-Factored: **the Coconut baseline does not encode Arousal or Dominance in its latent representations**, which directly explains its inferior performance on those dimensions relative to the CoT model.

6.4 Evaluation Metrics

All models are evaluated on the 861-sample test split using the following metrics:

- **Pearson** r per PAD dimension, measuring linear correlation between predicted and gold values.
- **Tolerance accuracy**: the proportion of test samples for which all three predicted dimensions fall within ± 0.25 of the gold values.
- **MAE** and **RMSE** over all three dimensions.

Predicted PAD values are extracted by parsing the model's generated output string.

6.5 Results

Table **??** summarises the evaluation results for all four models.

Model	Pearson V	Pearson A	Pearson D	Tol-Acc
CoT baseline	0.6858	0.4494	0.4548	46.92%
Coconut baseline	0.6841	0.3417	0.3363	47.74%
CoconutGPT-SimCoT	0.6768	0.2910	0.3261	42.86%
CoconutGPT-Factored	**0.8466**	**0.8119**	**0.8411**	45.30%

Table 6.2: Pearson correlation and tolerance accuracy on the EmoBank test set. CoconutGPT-Factored substantially outperforms all baselines on Arousal and Dominance.

CoconutGPT-Factored achieves a substantial improvement on Arousal (0.34 $\rightarrow$ 0.81) and Dominance (0.34 $\rightarrow$ 0.84), confirming that the VAD regression loss successfully drives the model to encode these dimensions. Valence also improves from 0.68 to 0.85. Tolerance accuracy (0.453) is slightly lower than the Coconut baseline (0.477), reflecting the fact that absolute numeric precision is a stricter criterion than linear correlation.

A follow-up linear probe on the CoconutGPT-Factored representations (Table **??**) reveals that, while Valence probe accuracy improves slightly ($r = 0.695 \rightarrow 0.732$), Arousal and Dominance probes remain weak ($r \approx 0.37$). This apparent contradiction – strong end-to-end performance but weak linear probe – indicates that the VAD information is encoded in a form that is accessible to the `vad_head` but not to a purely linear probe trained on question-only contexts. The discrepancy arises because the probe operates on latent vectors extracted before the answer prefix, whereas the `vad_head` at inference time has access to the full context including the answer preamble. This is an important limitation that motivates future work on aligning the latent space geometry with the PAD axes more directly.

Dimension	Pearson r (test)	R^2 (test)	R^2 (train)
Valence (V)	0.732	0.535	0.972
Arousal (A)	0.366	0.025	0.946
Dominance (D)	0.383	0.086	0.931

Table 6.3: Linear probe results on CoconutGPT-Factored latent representations. Despite strong end-to-end performance, Arousal and Dominance remain weakly decodable by a linear probe on question-only contexts.

6.6 Discussion

The results of this first experimental phase establish two key findings.

First, the linear probe confirms that the Coconut baseline's latent space does not encode Arousal or Dominance, which explains why it underperforms the CoT model despite having access to additional computation via continuous latent tokens. The latent tokens in the Coconut baseline appear to serve primarily as a computation budget for Valence-related reasoning, which is strongly correlated with surface-level lexical sentiment.

Second, the addition of direct VAD supervision via the regression and contrastive losses in CoconutGPT-Factored yields dramatic improvements on Arousal and Dominance, surpassing the CoT baseline by a large margin. This demonstrates that continuous latent space reasoning, when appropriately supervised, can encode richer affective structure than token-level generation alone.

Nevertheless, the weak linear probe results on the Factored model suggest that the improvement is driven by the `vad_head` decoding during generation, rather than by a fundamental reorganisation of the latent geometry. Achieving a latent space that is intrinsically aligned with PAD axes – such that the latent vectors themselves constitute interpretable, steerable affective representations – remains an open challenge and constitutes the primary direction for the next phase of this project.

6.7 Phase 2: Towards Steerable Latent Geometry

The linear probe results from Phase 1 establish a clear gap: although CoconutGPT-Factored achieves high end-to-end Pearson correlation (A: 0.81, D: 0.84), the Arousal and Dominance dimensions remain weakly decodable from the latent representations by a linear probe ($r \approx 0.37$). This matters because steering vectors are inherently linear interventions. If A and D are not linearly encoded in the latent space, a steering direction $\mathbf{d}_A$ derived from contrastive pairs cannot reliably shift Arousal without also disturbing Valence or semantic content. This chapter describes the architectural and training modifications introduced in Phase 2 to address this limitation.

6.7.1 Diagnosis: Why Does the Probe Fail?

The Phase 1 architecture uses a single shared regression head $\mathbf{W}_{\text{vad}} \in \mathbb{R}^{3\times d}$, whose three rows correspond to V, A, and D respectively. During optimisation, gradients from all three dimensions flow through the same weight matrix. Because Valence correlates strongly with surface lexical sentiment – a signal the model acquires readily – the shared head is dominated by the V gradient, leaving A and D directions weakly constrained. The contrastive loss similarly uses combined PAD-space Euclidean distance $\|\mathbf{v}_i^* - \mathbf{v}_j^*\|_2$, which conflates all three dimensions and further biases the latent geometry toward Valence.

A second contributing factor is the large gap between R^2_{train} and R^2_{test} (e.g. A: 0.946 vs. 0.025). This indicates that the linear probe overfits to training patterns rather than recovering a stable geometric direction in the latent space, consistent with the latent representations not having been explicitly regularised to place A and D along separable linear axes.

6.8 Architectural Changes

6.8.1 Separate Per-Dimension Projection Heads

The single shared head $\mathbf{W}_{\text{vad}}$ is replaced by three independent scalar projection heads:

$$\hat{V} = \mathbf{w}_V^\top \mathbf{z}, \quad \hat{A} = \mathbf{w}_A^\top \mathbf{z}, \quad \hat{D} = \mathbf{w}_D^\top \mathbf{z}$$

where $\mathbf{w}_V, \mathbf{w}_A, \mathbf{w}_D \in \mathbb{R}^d$ are fully independent parameter vectors and $\mathbf{z}$ is the mean-pooled latent representation. Each head receives its own gradient signal, preventing the V gradient from dominating the update to A and D directions.

6.8.2 Dimension-Weighted Regression Loss

The regression loss is reformulated as a weighted sum over dimensions:

$$\mathcal{L}_{\text{reg}} = \sum_{k \in \{V,A,D\}} w_k \cdot \frac{1}{B} \sum_{i=1}^{B} (\hat{k}_i - k_i^*)^2$$

with weights $(w_V, w_A, w_D) = (1.0, 2.0, 2.0)$, upweighting A and D to compensate for the natural advantage of Valence.

6.8.3 Per-Dimension Contrastive Loss

Rather than computing a single contrastive target from the combined PAD distance, the contrastive loss is decomposed into three independent terms, one per dimension:

$$s_{ij}^{*(k)} = \exp\left(-\frac{|k_i^* - k_j^*|}{\tau}\right), \quad k \in \{V, A, D\}$$

$$\mathcal{L}_{\text{con}} = \sum_{k \in \{V,A,D\}} w_k \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left(\cos(\mathbf{z}_i, \mathbf{z}_j) - s_{ij}^{*(k)}\right)^2$$

where $\mathcal{P}$ denotes all off-diagonal pairs in the mini-batch. This formulation pulls the latent space geometry toward each PAD axis independently, weighted by the same $(1.0, 2.0, 2.0)$ coefficients.

6.8.4 Orthogonality Regularisation

To encourage V, A, and D to occupy geometrically distinct directions in the latent space, an orthogonality penalty is imposed on the three head weight vectors:

$$\mathcal{L}_{\text{orth}} = \left\| \hat{\mathbf{W}} \hat{\mathbf{W}}^\top - \mathbf{I}_3 \right\|_F^2$$

where $\hat{\mathbf{W}} \in \mathbb{R}^{3\times d}$ stacks the ℓ_2-normalised rows $\hat{\mathbf{w}}_V, \hat{\mathbf{w}}_A, \hat{\mathbf{w}}_D$. This encourages the learned projection directions to be orthogonal, which is a necessary condition for applying steering vectors for individual PAD dimensions without cross-dimensional interference.

The full Phase 2 training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}} + \lambda_{\text{con}}\,\mathcal{L}_{\text{con}} + \lambda_{\text{orth}}\,\mathcal{L}_{\text{orth}}$$

with $\lambda_{\text{reg}} = 1.0$, $\lambda_{\text{con}} = 0.1$, $\tau = 1.0$, and $\lambda_{\text{orth}} = 0.1$.

6.9 Training Strategy: Staged Unfreezing

A further risk when warm-starting from an existing checkpoint is that the randomly initialised new heads produce large initial gradients that disturb the pre-trained latent representations before the heads have converged. To mitigate this, the base language model is frozen for the first $N_{\text{freeze}} = 3$ epochs, during which only the three projection heads are updated. After epoch 3, all parameters are unfrozen and trained jointly. This staged unfreezing allows the heads to align with the existing latent geometry before the backbone is permitted to reorganise in response to the affective supervision signal.

Chapter 7

Ethical Issues

This project investigates an emotional AI engine that operates on the basis of latent representations of emotions. It will utilise recent concept-based reasoning paradigms (e.g. latent and implicit inference models) and use human judgement to evaluate the output. While the project does not involve invasive experiments or high-risk applications, it raises several ethical and legal considerations.

7.1 Ethical Considerations

The main ethical considerations relate to the involvement of human participants in the evaluation phase. Human evaluators are expected to assess the outputs of the model rather than disclose personal experiences or emotional states. Participation should be voluntary, and they should be free to withdraw at any time without consequence. The data collected shall be kept to a minimum (e.g. numerical evaluations and brief qualitative judgments only) and no personally identifiable information is requested.

The second ethical consideration concerns the nature of the system under study. This project explores the expression and regulation of emotions in AI systems. There is a potential ethical risk that these technologies, if misused, could be perceived as enabling emotional manipulation and deceptive emotional signals. However, the manipulation techniques addressed in this study are essentially identical to the widely adopted usage patterns of LLMs today. For example, instructions such as "Please write more carefully" or "Could you rephrase this in a gentler manner?" are already routinely provided as prompts, and the models adjust the emotional tone of their output accordingly. In other words, the significance of this project lies not in endowing models with new capabilities for emotional manipulation, but in visualising the structures of emotional expression already present within the models. The target of manipulation is not human emotion but the latent representations within the model. The value of this research lies in treating the ambiguous tone adjustments already occurring as a subject of direct study, rather than leaving them as a black box.

7.2 Legal Considerations

The project will also use open source software, pre-trained models (e.g. GPT-2) and public datasets. These resources will be used strictly in accordance with their respective licences and for non-commercial academic research purposes only. No proprietary, restricted or export-controlled software or information will be generated.

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